CURRICULUM VITAE

Muhammad Umair Danish, PhD

Postdoctoral Fellow in Applied AI | **Western University**

Explainable AI • Mechanistic Interpretability • Human-AI Interaction

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158	5	5	19	1
CITATIONS	H-INDEX	I10-INDEX	PUBLISHED / ACCEPTED	BEST PAPER AWARD

Google Scholar metrics snapshot: 7 August 2026.

PROFILE

Academic profile

Postdoctoral Fellow in Applied AI at Western University whose research examines how machine-learning systems can be made more interpretable, reliable, and useful in applied settings. Research spans **mechanistic interpretability**, **human-centered explainable AI**, **physics-guided and temporal learning**, and **AI evaluation**, with peer-reviewed work across energy systems, financial decision support, computer vision, AI-generated image assessment, and research software. The program combines original model development with controlled baselines, ablations, statistical validation, visualization, reproducible implementation, and human-centered evaluation where appropriate.

APPOINTMENT

Current appointment

Postdoctoral Fellow in Applied AI — Jan 2026–Present

Western University, Faculty of Engineering — London, Ontario, Canada

Research on mechanistic interpretability and human-centered explainable AI under the supervision of Dr. Katarina Grolinger, with emphasis on faithful non-interventional analysis of neural representations, controlled experimental design, statistical validation, reproducibility, and publication-grade technical evidence.

EDUCATION

Education

PhD, Electrical and Computer Engineering — Dec 2025

Western University — London, Ontario, Canada

Specialization: Applied Artificial Intelligence and Explainable AI
Dissertation: *Neural Networks and Their Interpretability for Building Energy Modeling*
Advisor: Dr. Katarina Grolinger

MS, Software Engineering — Dec 2019

University of Lahore — Pakistan

Thesis: *Image Denoising and Restoration Using Convolutional Neural Networks* **Advisor:** Dr. Muhammad Sultan Zia

BS, Software Engineering — 2016

COMSATS University Islamabad — Pakistan

Capstone: *A Comparative Analysis of Image Restoration Techniques* **Advisor:** Dr. Muzammil Behzad

SCHOLARSHIP

Peer-reviewed journal publications

Published and accepted works are listed in reverse chronological order. The candidate's name is shown in bold.

J1 Demian Kogutek, Laura Carroll, **Muhammad Umair Danish**, and Cathy Lu. "Dispersity Measures Within Sessions During Improvised Active Music Therapy with Clients with Parkinson's Disease." *Approaches: An Interdisciplinary Journal of Music Therapy*.
Accepted for publication, 27 July 2026; copyediting pending.

J2 D. Kogutek and **M. U. Danish**. "MidiPy: A Python Application for Processing MIDI Data in Improvised Active Music Therapy." *Journal of Creative Music Systems*.
Accepted, 2026. Code repository.

J3 M. Buwaneswaran, **M. U. Danish**, and K. Grolinger. "Conditional Denoising Diffusion Model for Missing Value Imputation in Smart Meter Data." *Engineering Applications of Artificial Intelligence*, Vol. 181, Article 115403.
Published, 2026.

J4 **M. U. Danish**, M. Aziz, U. Rehman, and K. Grolinger. "Linear Lens: A Human-Centered, Non-Interventional Mechanistic Approach to Explainable AI." *Machine Learning with Applications*, Vol. 24, Article 100897.
Published, 2026. doi:10.1016/j.mlwa.2026.100897 | Code.

J5 **M. U. Danish**, U. Rehman, and K. Grolinger. "Monotone Delta: An Order-Theoretic Tournament Graph Approach for Internal Consistency Assessment." *Frontiers in Applied Mathematics and Statistics*, Vol. 12, Article 1687194.
Published, 2026. doi:10.3389/fams.2026.1687194.

J6 M. Aziz, U. Rehman, **M. U. Danish**, and K. Grolinger. "Global-Local Image Perceptual Score (GLIPS): Evaluating Photorealistic Quality of AI-Generated Images." *IEEE Transactions on Human-Machine Systems*, Vol. 55, No. 2, pp. 223–233.
Published, 2025. doi:10.1109/THMS.2025.3527397 | Code.

J7 **M. U. Danish** and K. Grolinger. "Kolmogorov-Arnold Recurrent Network for Short-Term Load Forecasting Across Diverse Consumers." *Energy Reports*, Vol. 13, pp. 713–727.
Published, 2025. doi:10.1016/j.egyr.2024.12.038 | Code.

J8 **M. U. Danish**, K. Ali, K. Siddiqui, and K. Grolinger. "Physics-Guided Memory Network for Building Energy Modeling." *Energy & AI*, Vol. 21, Article 100538.
Published, 2025. doi:10.1016/j.egyai.2025.100538 | Code.

J9 M. Aziz, U. Rehman, **M. U. Danish**, S. Ali, and A. Z. Abbasi. "Towards a Unified Evaluation Framework: Integrating Human Perception and Metrics for AI-Generated Images." *Multimedia Systems*, Vol. 31, No. 3, Article 180.
Published, 2025. doi:10.1007/s00530-025-01769-7.

J10 **M. U. Danish** and K. Grolinger. "Leveraging Hypernetworks and Learnable Kernels for Consumer Energy Forecasting Across Diverse Consumer Types." *IEEE Transactions on Power Delivery*, Vol. 40, No. 1, pp. 75–87.
Published, 2024. doi:10.1109/TPWRD.2024.3486010 | Code.

CONFERENCES

Peer-reviewed conference publications

C1 **M. U. Danish**, M. Aziz, U. Rehman, and K. Grolinger. “Cognitive Effects of Textual, Graphical, and Interactive Explanations in Explainable Artificial Intelligence.” *52nd Annual Conference of the IEEE Industrial Electronics Society (IECON 2026)*.
Accepted for presentation, 2026.

C2 **M. U. Danish** et al. “Human-Centered Explainable Artificial Intelligence and the Effects of Explanation Modalities on Trust, Understanding, and Decision Making.” *52nd Annual Conference of the IEEE Industrial Electronics Society (IECON 2026)*.
Accepted for presentation, 2026; final bibliographic record pending.

C3 A. A. Yousaf, M. Aziz, **M. U. Danish**, S. A. Safi, and U. Rehman. “Statistically-Guided Kernel Regression: Embedding Feature Relevance into Kernel Learning.” *2nd International Conference on Advanced Machine Learning and Data Science*.
Accepted for oral presentation, Osaka, Japan, 2026.

C4 D. Tissera, O. Awadallah, **M. U. Danish**, A. Sadhu, and K. Grolinger. “Any-Class Presence Likelihood for Robust Multi-Label Classification with Abundant Negative Data.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
Published, 2026. CVPR record.

C5 M. Aziz, **M. U. Danish**, and U. Rehman. “Distributional Feature Separability for Explainability of Neural Networks for Financial Credit Evaluation.” *IEEE Conference on AI, Science, Engineering, and Technology (AIxSET)*, pp. 127–133.
Published, 2025. **Best Paper Award**. doi:10.1109/AIXSET65682.2025.00024.

C6 M. Aziz, **M. U. Danish**, F. Iqbal, and U. Rehman. “Finite-Difference Approximation for Explaining Neural Network Predictions in Financial Credit Evaluation.” *International Conference on Advanced Machine Learning and Data Science (AMLDS)*.
Published, Tokyo, Japan, 2025. doi:10.1109/AMLDS63918.2025.11159472.

C7 **M. U. Danish**, A. A. Yousaf, M. Aziz, K. Grolinger, and U. Rehman. “RootSolver: A Cognitive-Inspired Approach to Polynomial Root-Finding with Multiplicity Detection.” *17th International Conference on Advanced Computational Intelligence*.
Published, 2025. doi:10.1109/ICACI65340.2025.11096224.

C8 **M. U. Danish**, M. Noorchenarboo, A. Narayan, and K. Grolinger. “ChebyRegNet: An Unsupervised Deep Learning Technique for Deformable Medical Image Registration.” *50th Annual Conference of the IEEE Industrial Electronics Society (IECON)*.
Published, 2024. doi:10.1109/IECON55916.2024.10905705.

C9 **M. U. Danish**, M. Buwaneswaran, T. Fonseka, and K. Grolinger. “Graph Attention Convolutional U-Net: A Semantic Segmentation Model for Identifying Flooded Areas.” *50th Annual Conference of the IEEE Industrial Electronics Society (IECON)*.
Published, 2024. doi:10.1109/IECON55916.2024.10905482.

MANUSCRIPTS

Manuscripts under review

M1 **M. U. Danish**, U. Rehman, and K. Grolinger. “Understanding the Inner Workings of Memory Networks with Multi-Stage Mechanistic Interpretability.”
In revision at *IEEE Transactions on Artificial Intelligence*.

M2 **M. U. Danish**, U. Rehman, and K. Grolinger. “Behavioral Mechanistic Interpretability for Enhancing Human Cognition in Explainable AI.”
Under review at *IEEE Transactions on Human-Machine Systems*.

M3 **M. U. Danish**, U. Rehman, and K. Grolinger. “AI-SSIM: Human-Centric Image Assessment Through Pseudo-Reference Generation.”
Under review at *Computer Vision and Image Understanding*.

EXPERIENCE

Research and professional appointments

Postdoctoral Fellow in Applied AI
Jan 2026–Present

Western University London, Ontario, Canada

- Lead research on mechanistic interpretability and concept-based explainability, developing faithful, non-interventional methods for understanding neural representations and decision pathways.
- Design studies spanning human-centered XAI, model reliability, internal-representation analysis, and evaluation of AI-generated content.
- Build reproducible PyTorch research pipelines with controlled baselines, ablations, statistical validation, visualization, and publication-ready reporting.

Graduate Research Assistant
2023–2025

Western University London, Ontario, Canada

- Developed neural forecasting methods for building energy and heterogeneous consumer loads, including physics-guided memory, hypernetwork/kernel, and Kolmogorov-Arnold recurrent architectures.
- Built end-to-end experimental workflows for preprocessing, model training, validation, cross-consumer generalization, interpretability, performance comparison, and failure analysis.
- Translated applied research into first-author journal publications in *Energy & AI*, *IEEE Transactions on Power Delivery*, and *Energy Reports*.

Research Assistant, Part-Time
2023–Present

Wilfrid Laurier University Waterloo, Ontario, Canada

- Co-developed `MidiPy`, an open-source Python toolkit that transforms MIDI event streams into reproducible measures and visual analytics for improvised active music-therapy research.
- Contributed software design, data-processing workflows, documentation, packaging, usability, and translation for interdisciplinary health-science collaborators.

Research Assistant
Jun 2022–Dec 2022

Gwangju Institute of Science and Technology (GIST) South Korea

- Implemented and evaluated deep-learning workflows for biomedical image restoration, deformable registration, and segmentation, emphasizing controlled preprocessing and reliable evaluation.

Lecturer
2019–2021

University of Lahore Pakistan

- Taught machine learning, statistics, and software-engineering topics; designed assessments aligned with learning outcomes and supervised applied student projects.
- Mentored students in implementation strategy, experimental reasoning, technical communication, and project delivery.

CONTRIBUTIONS

Selected research contributions

Mechanistic interpretability. Developed Linear Lens for non-interventional tracing of predictive information through neural networks; current work extends this direction to multi-stage and behavioral mechanistic analysis.

Human-centered AI evaluation. Connected computational evidence with human outcomes through explanation-modality studies, perceptual image-quality assessment, and internal-consistency analysis.

Energy and temporal intelligence. Developed physics-guided memory, hypernetwork/kernel, and Kolmogorov-Arnold recurrent methods for energy forecasting, alongside diffusion-based smart-meter imputation.

Vision and interdisciplinary AI. Contributed to robust multi-label classification, flooded-area segmentation, medical image registration, financial decision support, and music-therapy analytics.

SOFTWARE

Research software and open-source work

MidiPy - Python research software — 2023–Present

Wilfrid Laurier University collaboration — Open source

Python software for processing MIDI data in improvised active music-therapy research, supporting reproducible data transformation and visual analytics.

PyPI | GitHub | associated journal article accepted in *Journal of Creative Music Systems*.

TEACHING

Teaching and mentoring

Teaching Assistant — 2023–2025

Western University — Faculty of Engineering

Courses: ES1036 Programming Fundamentals for Engineers; SE3309 Database Management Systems; SE2251 Software Design; SE3350 Software Engineering Design I.

Supported programming, databases, software design, debugging, engineering problem solving, assessment interpretation, and code quality.

Lead Teaching Assistant - SE3309 — Sep 2024–Dec 2024

Western University — Database Management Systems

Coordinated TA workflows, grading alignment, structured rubrics, solution guides, review materials, and teaching-team communication.

METHODS

Technical profile

Programming	Python, MATLAB, Bash, SQL/MySQL, LaTeX; R (basic).
ML / Deep Learning	PyTorch, TensorFlow/Keras, neural networks, CNNs, RNNs, GNNs, diffusion models, time-series forecasting, representation learning, kernel methods, physics-guided learning.
Interpretability	Mechanistic interpretability, concept-based explainability, finite-difference explanation, attribution and diagnostic analysis, representation visualization, model auditing, human-centered XAI evaluation.
Research workflow	Experimental design, controlled baselines, ablation studies, statistical analysis, visualization, experiment tracking, reproducible implementation, manuscript preparation, publication-quality figures.
Engineering tools	Git/GitHub, Linux, Jupyter, Conda, packaging and release workflows including PyPI.

RECOGNITION

Honors and awards

Western Postdoctoral Fellowship, Western University	2026
Best Paper Award, IEEE Conference on AI, Science, Engineering, and Technology (AIxSET)	2025
Western Graduate Fellowship, Western University	2024
Western Graduate Research Scholarship, Western University	2023
Korean Government Scholarship	2022

SERVICE

Scholarly service

Peer reviewer: *Engineering Applications of Artificial Intelligence*; *IEEE Transactions on Industrial Informatics*; *IEEE Transactions on Artificial Intelligence*; *Energy & AI*; *Applied Energy*.

Publication status and bibliographic details reflect the CV and website records available on 8 August 2026. Citation metrics are dynamic.